

Technology Stack

Project Name: CaptionAI: AI-powered social media caption generator

Date: 7 July 2026

Team Name: Anupriya

Maximum Marks: 2 Marks

Technology Stack Details

S.No	Architecture Component/Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React.js & Tailwind CSS	React allows for reusable UI components to easily build clean text containers, while Tailwind provides responsive utilities to format the layout smoothly across mobile viewports.
2	Backend / Server-Side	Node.js (Express.js)	Offers lightning-fast asynchronous processing and non-blocking I/O operations, which is ideal for handling multi-tone prompt payload streams without lagging server performance.
3	Database / Data Storage	MongoDB	A flexible, document-oriented NoSQL database that cleanly handles unstructured user generation history logs and metadata properties without strict schema overhead.
4	Cloud / Hosting / Deployment	Amazon Web Services (AWS) & Docker	Docker containerizes the application modules for environment consistency, while AWS (ECS/Fargate) ensures seamless vertical scalability to handle influxes of active concurrent sessions.

S.No	Architecture Component/Layer	Technology Chosen	Justification / Purpose
5	Version Control & CI/CD	GitHub & GitHub Actions	Enables effortless Git collaboration for tracking codebase variants and automates unit test deployment scripts to maintain stable integration pipelines.
6	Third-Party APIs / Other Tools	OpenAI API & Stripe API	The OpenAI API acts as the core LLM engine providing the intelligent caption alternatives, while Stripe processes premium billing operations securely.